

Project Title

CA PracticeOS Compliance — AI Agent Layer

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Name / Membership Details

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Problem Statement

A CA firm's practice-management work is document-heavy and largely manual: client details get retyped from PAN/GST documents, engagement letters and certificates are drafted from scratch each time, and checking a drafted document against the mandatory clauses a professional standard requires is a manual, easy-to-skip step. The Level 1 version of this project (CA PracticeOS Compliance) built a browser-based document engine, but it had no reasoning layer, no real login/access control, and existed only as a file on one person's computer — not something a firm's staff could actually log into and use together.

Objective

To add a real, multi-step reasoning AI agent layer directly into the existing L1 application, and to turn it into a genuine, commercially usable multi-firm SaaS product — not a local prototype and not a second, separate demo app — so that a CA firm can onboard clients faster, draft documents from plain English, and have every drafted document automatically checked against mandatory professional clauses before it goes out the door.

AI Technology Used

Claude API (model: claude-sonnet-5), called from a Python/FastAPI backend, one call per agent reasoning step, chained together rather than a single one-shot prompt:

- Agent 3 — Compliance Checklist Agent (headline feature).** A perceive → reason → self-check → act loop: one Claude call verdicts every mandatory clause with quoted evidence from the document text, then an independent self-check pass catches the model over-claiming a clause is present, before a final pass/fail verdict is produced.
- Agent 1 — Client Onboarding Agent.** Claude's vision capability reads a photo/PDF of a PAN card or GST certificate and extracts structured, validated fields (name, PAN, GSTIN, address), cross-checking that the GSTIN embeds the same PAN.
- Agent 2 — Document Drafting Agent.** Claude extracts client name, assignment type, fees, and other parameters from one plain-English instruction, then fills the app's own document template — its output is fed straight into Agent 3.

How the Solution Works

The L1 frontend (a single HTML file, installable as a PWA) is hosted publicly on GitHub Pages. It calls a FastAPI backend deployed on Render for every AI agent call and every login/team-management action; the backend talks to Claude for reasoning and to a permanent Supabase Postgres database for all tenant, user, and client data. A firm signs up through the app, the platform owner approves them by email (via the Resend API), and from then on that firm's admin can log in, add their own staff with roles (Preparer/Reviewer/Approver/Admin), and use the full document suite — completely isolated from every other firm on the same app. The Tally Connector talks directly from the browser to Tally Prime's own local XML/HTTP gateway via a small CORS relay, for real live ledger and company data, with an honestly-disclosed limitation that this specific piece only works on the accountant's own machine (see Learnings/Future Scope).

Layer	Implementation
Frontend	CA_PracticeOS_Compliance.html — single-file, installable PWA, hosted on GitHub Pages
Backend	Python 3.11, FastAPI — auth, team management, 3 agent endpoints, deployed on Render

Database	Supabase Postgres (persistent, multi-tenant: Tenants / Users / Clients)
LLM	Claude API (claude-sonnet-5) — one call per agent step, chained
Email	Resend API — admin-approval notifications on signup
Connector	Direct browser → local CORS relay → Tally Prime XML/HTTP gateway

Live app: cashubhammishra-ctrl.github.io/CA_PracticeOS_Compliance/CA_PracticeOS_Compliance.html | Backend: ca-practiceos-backend.onrender.com (see [deployment_link.txt](#) for full detail)

Key Features

- AI Compliance Agent that checks a drafted document against mandatory clauses and reports exactly what's missing, with quoted evidence for what's present.
- AI Onboarding Agent that turns a PAN/GST photo into a validated Client Master row in one step.
- AI Drafting Agent that turns one plain-English line into a filled, ready-to-check document draft.
- Live Tally Prime connector for real company and ledger data.
- Full multi-tenant SaaS layer: signup with admin-approval-by-email, JWT login, per-firm data isolation, a Team Members page for a firm's own admin to add/revoke/restore staff by role, and a separate platform-owner control page to revoke or restore an entire firm's access at any time.
- Full document suite carried over and extended from L1: engagement letters, certificates, net worth certificates, fee notes, working papers, checklists, UDIN register, GST engine, and more.
- Installable Progressive Web App, responsive across desktop, tablet, and mobile.

Results / Benefits

- Agent 3 caught a genuine gap in the app's own engagement letter template — not a synthetic test case. Feeding Agent 2's draft into Agent 3 produces a real FAIL for a missing Limitations clause, because that clause was never in the original template — proving the compliance check finds real gaps, not just staged ones.
- Full auth flow verified end to end against the live public URL, the deployed Render backend, and the permanent Supabase database: signup → admin-approval email → approve → login → JWT session → tenant data isolation between separate test firms → role-based team permissions (a non-admin blocked from adding members, 403 confirmed).
- Revoke/Restore verified live: a revoked team member is immediately blocked from logging in (403), and restoring them immediately allows login again (fresh JWT issued).
- Responsive design verified across desktop, tablet, and mobile viewports — off-canvas navigation, full-width modals, horizontally-scrollable tables, no layout breakage.
- Net benefit to a real CA firm: less manual retyping, faster first drafts, and a genuine safety net against sending out a document that's missing a mandatory clause — while giving the firm real control over who on their team can access the system.

Future Scope

Zoho Books and SAP connectors via the same connector interface. A local sync-agent for Tally that removes the need for the browser-direct connection — the agent would talk to Tally locally and push data to the cloud backend, so the web app itself never has to touch Tally directly (see Learnings below). A password-reset flow for user accounts. Server-side audit logging for every SaaS action (currently client-side only for local-use document features).

Learnings

A validating agent needs a genuinely independent second opinion, not just a JSON wrapper around one model call. Verifying a third-party integration's real architecture before building around it matters: Tally Prime's gateway is a local-only HTTP server that sends no CORS headers, so a browser refuses to read its response without a small relay in between — even though modern browsers do allow an HTTPS page to reach a local server at all. Rather than hide this constraint, the app discloses it honestly. The same build-then-verify discipline applied to the SaaS layer: every claim above (isolation, revoke, responsive layout) was tested live against the deployed system, not assumed from the code alone.